



ENGINEERING MATHEMATICS-II

MATLAB

Department of Science and Humanities

Fourier Series

The Fourier series expansion of a function $f(x)$ over the interval $(\alpha, \alpha + 2\pi)$

is given by
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx],$$

where,
$$a_0 = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) dx;$$

$$a_k = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cos kx dx; k = 1, 2, \dots$$

$$b_k = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \sin kx dx; k = 1, 2, \dots$$

Problem

Find the Fourier series for the function $f(x) = x - x^2$ from $x = -\pi$ to $x = \pi$.

```
>> syms x k n pi
```

```
>> evalin(symengine,'assume(k,Type::Integer)');
```

 The `evalin` command tells
MATLAB that k is an integer.

```
>> a = @(f,x,k,pi) int(f*cos(k*x)/pi,x,-pi,pi);
```

 k th Fourier cosine coefficient of f

```
>> b = @(f,x,k,pi) int(f*sin(k*x)/pi,x,-pi,pi);
```

 k th Fourier sine coefficient of f

```
>> f=x-x^2
```

```
>> A=simplify(a(f,x,k,pi))
```

```
-(2*(k^2*pi^2*sin(k*pi) - 2*sin(k*pi) + 2*k*pi*cos(k*pi)))/(k^3*pi)
```

```
>> pretty(A)
```

$$\frac{(k^2 \pi^2 \sin(k \pi) - 2 \sin(k \pi) + 2 k \pi \cos(k \pi))^2}{k^3 \pi^3}$$

```
>> B=simplify(b(f,x,k,pi))
```

$$B = (2 * (\sin(k * \pi) - k * \pi * \cos(k * \pi))) / (k^2 * \pi)$$

```
>> pretty(B)
```

$$\frac{2 (\sin(k \pi) - k \pi \cos(k \pi))}{k^2 \pi}$$

The nth partial sum

```
>> fs = @(f,x,n,pi) a(f,x,0,pi)/2 + ...
        symsum(a(f,x,k,pi)*cos(k*x) + b(f,x,k,pi)*sin(k*x),k,pi,n);
```

```
>> pretty(fs(f,x,3,pi))
```

$$\frac{\left(\frac{\cos(kx) (k^2 \pi^2 \#1 - 2 \#1 + 2 k \pi \#2)}{k^3} \right) - \left(\frac{\sin(kx) (\#1 - k \pi \#2)}{k^2} \right)}{\pi^3}$$

where

```
#1 == sin(k pi)
```

```
#2 == cos(k pi)
```

Problem



Find the Fourier series for the function $f(x) = x^2$ from $x = -\pi$ to $x = \pi$.

```
>> syms x k n pi
```

```
>> evalin(symengine,'assume(k,Type::Integer)');
```

```
>> a = @(f,x,k,pi) int(f*cos(k*x)/pi,x,-pi,pi);
```

```
>> b = @(f,x,k,pi) int(f*sin(k*x)/pi,x,-pi,pi);
```

```
>> f=x^2
```

```
>> A=simplify(a(f,x,k,pi))
```

```
(2*(k^2*pi^2*sin(k*pi) - 2*sin(k*pi) + 2*k*pi*cos(k*pi)))/(k^3*pi)
```

```
>> pretty(A)
```

$$\frac{(k^2 \pi^2 \sin(k \pi) - 2 \sin(k \pi) + 2 k \pi \cos(k \pi))^2}{k^3 \pi}$$

```
>> B=simplify(b(f,x,k,pi))
```

B = 0

```
>> fs = @(f,x,n,pi) a(f,x,0,pi)/2 + ...
        symsum(a(f,x,k,pi)*cos(k*x) + b(f,x,k,pi)*sin(k*x),k,pi,n);
```

```
>> pretty(fs(f,x,2,pi))
```

$$\frac{\pi^2}{3} + \frac{\cos(kx) (k^2 \pi^2 \sin(k\pi) - 2 \sin(k\pi) + 2 k \pi \cos(k\pi))}{k^3 \pi^2}$$

Fourier Series over an arbitrary interval

The Fourier series expansion of a function $f(x)$ over the interval $(-L, L)$ is

given by $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$, where

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx;$$

$$a_k = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{k\pi x}{L}\right) dx; k = 1, 2, \dots$$

$$b_k = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{k\pi x}{L}\right) dx; k = 1, 2, \dots$$

Problem



Find the Fourier series for the function $f(x) = |x|$ from $x = -1$ to $x = 1$.

```
>> syms x k n pi
```

```
>> evalin(symengine,'assume(k,Type::Integer)');
```

```
>> a = @(f,x,k,L) int(f*cos(k*pi*x/L)/L,x,-L,L);
```

```
>> b = @(f,x,k,L) int(f*sin(k*pi*x/L)/L,x,-L,L);
```

```
>> f=abs(x)
```

```
>> A=simplify(a(f,x,k,1))
```

$$-(2*(2*\sin((k*\pi)/2)^2 - k*\pi*\sin(k*\pi)))/(k^2*\pi^2)$$

```
>> pretty(A)
```

$$\frac{\left| \sin \left(\frac{k \pi}{2} \right) \right|^2 - \sin^2(k \pi)}{k^2 \pi^2}$$

```
>> B=simplify(b(f,x,k,1))
```

B = 0

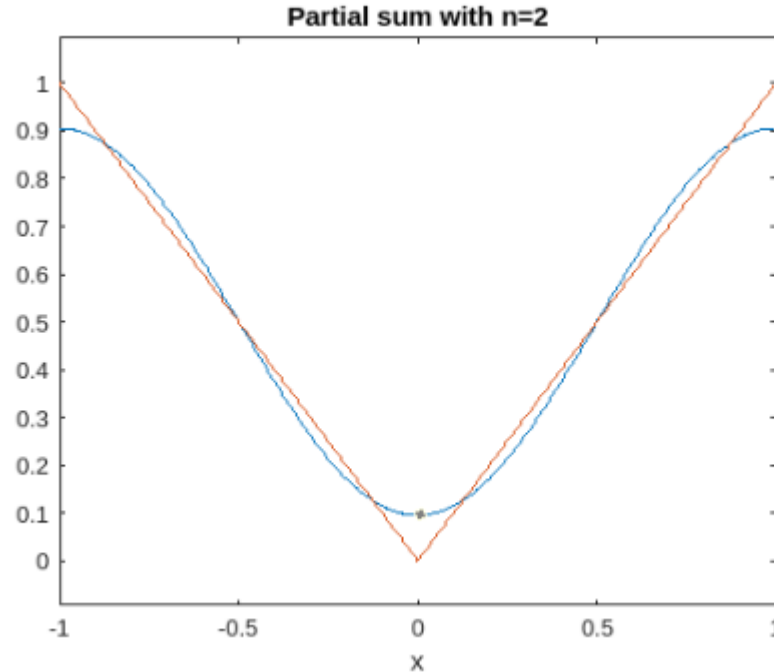
```
>> fs = @(f,x,n,L) a(f,x,0,L)/2 + ...
    symsum(a(f,x,k,L)*cos(k*pi*x/L) + b(f,x,k,L)*sin(k*pi*x/L),k,1,n);
```

```
>> pretty(fs(f,x,2,1))
```

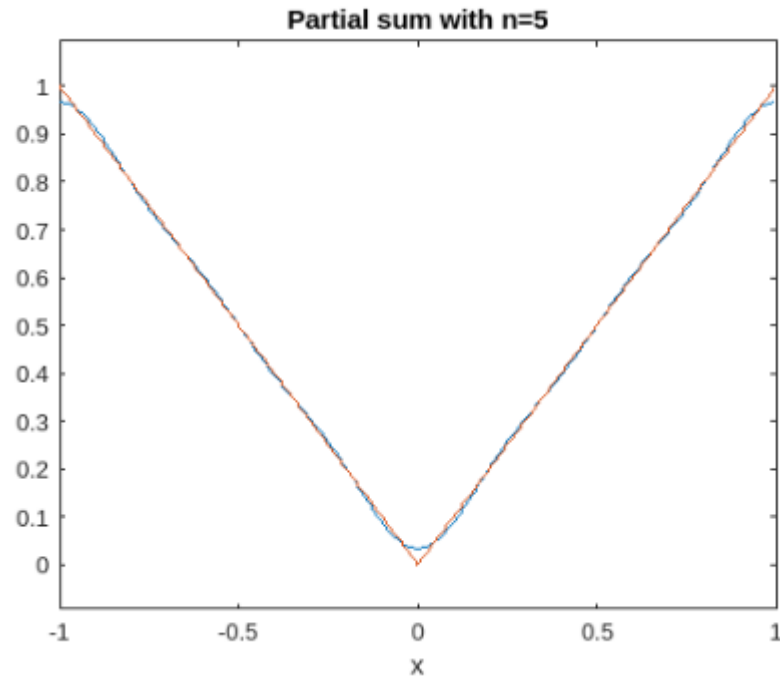
$$\frac{\cos(2\pi x) (\pi \sin(2\pi)^2 - 2 \sin(\pi)^2) + \frac{\cos(\pi x) \sqrt{\sin^2(\pi) - 2} - \pi \sin(\pi) \sqrt{2}}{2}}{2\pi} + \frac{1}{2}$$

Here are the plots of the partial sums for $n=2,5,10$. The plot also shows the function f .

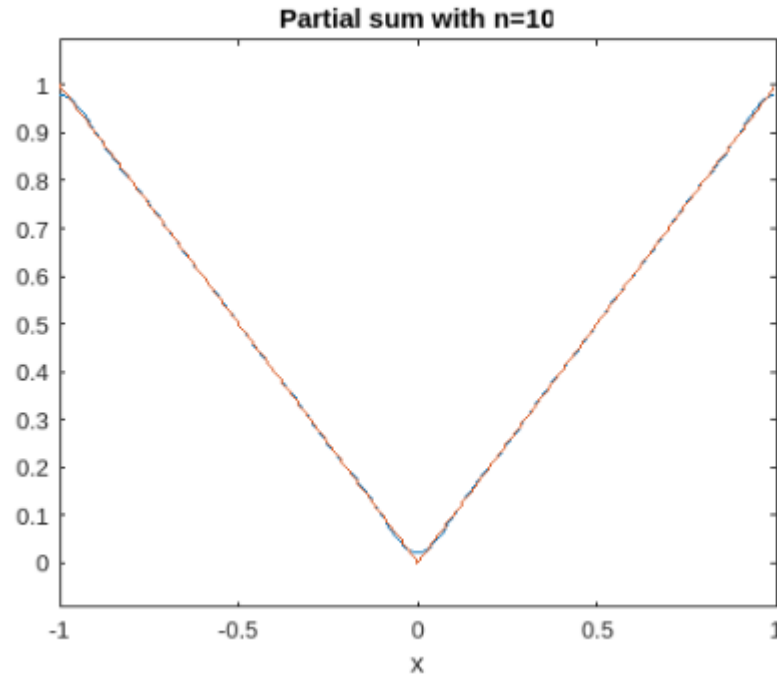
```
>> ezplot(fs(f,x,2,1),-1,1)
>> hold on
>> ezplot(f,-1,1)
>> hold off
>> title('Partial sum with n=2')
```



```
>> ezplot(fs(f,x,5,1),-1,1)
>> hold on
>> ezplot(f,-1,1)
>> hold off
>> title('Partial sum with n=5')
```



```
>> ezplot(fs(f,x,10,1),-1,1)
>> hold on
>> ezplot(f,-1,1)
>> hold off
>> title('Partial sum with n=10')
```





THANK YOU
